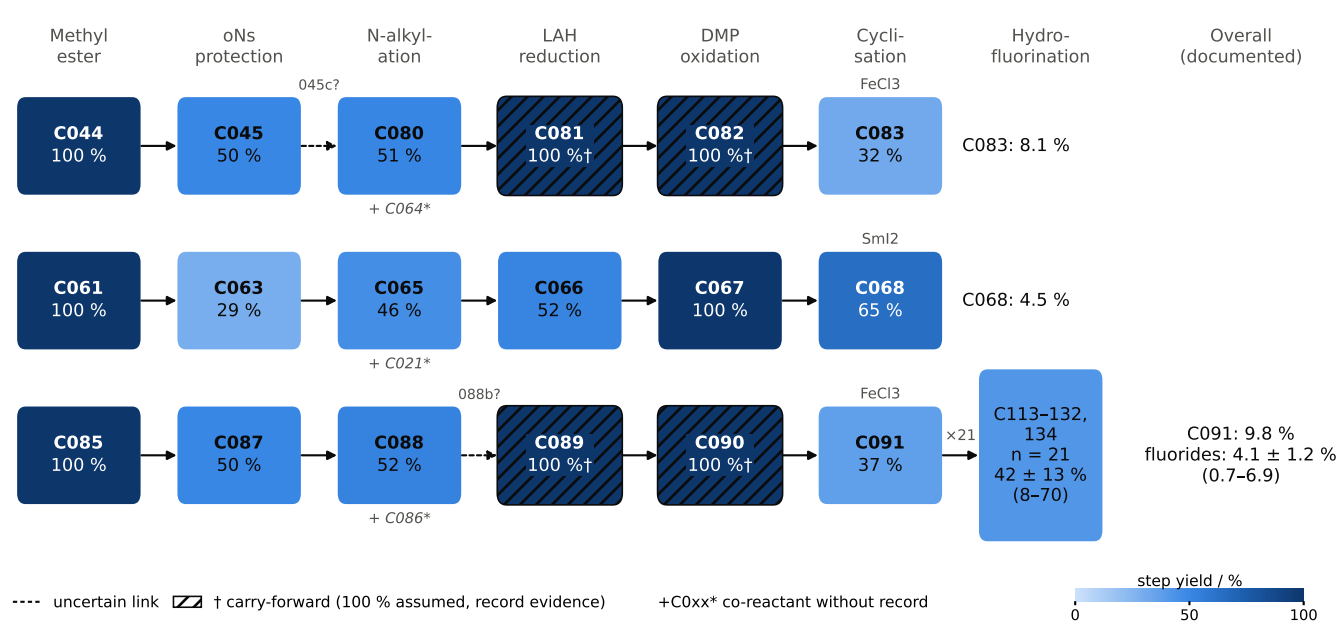
Methyl ester
oNs protection
N-alkyl-ation
LAH reduction
DMP oxidation
Cycli-sation
Hydro-fluorination
Overall (documented)
C044 100 %
C045 50 %
045c?
C080 51 %
+ C064*
C081 100 %†
C082 100 %†
FeCl3
C083 32 %
C083: 8.1 %
C061 100 %
C063 29 %
C065 46 %
+ C021*
C066 52 %
C067 100 %
SmI2
C068 65 %
C068: 4.5 %
C085 100 %
C087 50 %
C088 52 %
+ C086*
088b?
C089 100 %†
C090 100 %†
FeCl3
C091 37 %
×21
C113–132, 134
n = 21
42 ± 13 %
(8–70)
C091: 9.8 %
fluorides: 4.1 ± 1.2 %
(0.7–6.9)
uncertain link
† carry-forward (100 % assumed, record evidence)
+C0xx* co-reactant without record
step yield / %
0
50
100